

The Seen World Beyond The Stargate

The central visual pathway — from the Stargate to the seer

Alexandru C. Stan

© 2026 Alexandru C. Stan

The Seen World Beyond The Stargate — the central visual pathway.

Companion to Vision: From Photon to Mind; the two publish as a pair.

Set in STIX Two Text. Typeset in the Lapidary style.

Contents

1. Through the Stargate: the optic nerve and the chiasm.....	1
Through the Stargate.....	1
The chiasm: the first crossing.....	2
Reading the wiring backward.....	2
When the instrument authors the observation.....	3
Light left at the door.....	4
2. The lateral geniculate nucleus: the gate that mostly listens to the brain.....	5
The little knee.....	5
Six layers, still one eye at a time.....	5
Counting the synapses.....	6
Driver and modulator.....	7
The gate that expects.....	8
The stale present.....	8
3. V1: the cortex that finds the edges.....	10
From the dot to the line.....	10
The two eyes, fused at last.....	11
Columns, hypercolumns, and the magnified map.....	12
The mathematics of the edge.....	12
The edge that isn't there.....	13
4. The two streams: where seeing becomes knowing and acting.....	14
The fork.....	14
Knowing without doing, doing without knowing.....	15
Rebuilding the frame.....	15
The map with no viewer.....	16
What the streams leave us.....	16
5. Coda: the authored world and the seer.....	18
The world with the light off.....	18
The shadow and the fixed point.....	18
Why a world, and not a proof.....	19
Forms all the way down?.....	20
The seer who could be otherwise.....	21
The seer.....	21
References.....	23

Figures

1. The visual pathway and the geometry of its lesions.....	3
2. What lands on a geniculate relay cell.....	6
3. From the dot to the oriented edge.....	11
4. The two cortical streams.....	14
5. The ladder of mediation.....	19

*The standalone companion to **Vision: From Photon to Mind**. The first volume followed the light to the back of the eye and left the signal, now electrochemical, at the threshold of the brain. This volume steps through that threshold and follows the seeing. A matched pair: the first volume's coda points forward; this one's opening points back.*

1. Through the Stargate: the optic nerve and the chiasm

Through the Stargate

The first volume stopped, deliberately, at a threshold — the last structure that is still the eye — and left the seeing to begin on the far side. So begin at the seam. A photoreceptor has fallen quiet; bipolar and ganglion cells have split the message into ON and OFF, centre and surround, fast and slow; and the last cell of the retina has fired the result as a train of spikes down its own axon. That axon now reaches the wall the first volume named the Stargate — the optic disc, where somewhat more than a million axons converge and pass together through the *lamina cribrosa*, the sieve of collagen in the sclera where the retina stops and the nerve begins. On the near side lay optics and photochemistry: refraction, absorption, transduction. We step through. On the far side the rules change — the subject is no longer how light is *caught* but how a brain *makes a seen world* out of spike trains — and the change is not only figurative. It happens in the tissue, in the space of a few microns, exactly here.

Inside the eye the axons run *bare*. They must. Myelin is opaque — a fatty, white, light-scattering sheath — and any myelinated wire laid across the retina would cast its shadow on the very surface it serves; so the eye leaves its inner wiring unclad, paying in conduction speed for the sake of a clear window.^[1] The instant an axon passes through the lamina cribrosa, that economy reverses: oligodendrocytes wrap it in myelin — oligodendrocytes, the insulators of the *brain*, not the Schwann cells that sheathe every ordinary nerve in the body.^[1] At that pore, retina becomes central nervous system. The "optic nerve" is misnamed. It is not a peripheral nerve at all but a tract of white matter — a length of brain extruded down the eye socket — and embryologically it never was anything else: the eye buds from the diencephalon and creeps forward to meet the light, trailing its cable behind it.^[1]

The consequences are not academic; they are what a physician reads at the bedside. Because the tract is brain, it demyelinate like brain — multiple sclerosis, which spares true peripheral nerves, strips the optic nerve, and an attack of optic neuritis is often the disease's first word.^[2] Because the tract is brain, it is wrapped in the same three meninges and bathed in the same cerebrospinal fluid; when pressure rises inside the skull it travels down this

sleeve and swells the nerve head, so that the swollen disc — papilledema — becomes a window through which the pressure inside a living skull is *seen*, through the pupil, by a beam of light.^[2] The optic nerve is the only tract of the central nervous system a clinician can look upon directly.^[2] The first volume was spent learning how the eye admits light; at the threshold, we turn the light back on the brain, to look at the brain.

The chiasm: the first crossing

A few centimetres in, the two nerves meet and do something stranger than merely touch: they partly cross. At the *optic chiasm* — named for the Greek letter X, the crossing — the fibres from the nasal half of each retina decussate to the opposite side, while the fibres from the temporal half stay home. The split is not even. A little over half cross and a little under half do not: in the classic human count, some fifty-three fibres cross for every forty-seven that remain, out of somewhat more than a million in each nerve.^[3]

The first volume's coda named this crossing and set it aside for this one; here is the idea it was keeping. The purpose of the half-crossing is the first deep idea of the central pathway, and it is easy to state wrongly. It is *not* that the left eye reports to the right brain — that is the plan of a creature with eyes on the sides of its head, each guarding its own separate world. We are not built that way. Our eyes face forward and overlap; each sees, from a slightly shifted vantage, the *same* scene. What the chiasm sorts, therefore, is not eyes but *fields*. Because the lens inverts and reverses the image — the optics of the first volume — the right half of the visual world falls on the nasal retina of the right eye and the temporal retina of the left; the chiasm gathers both of those and routes them, together, to the left hemisphere. Each half of the brain is handed the *opposite half of the world*, assembled from both eyes at once.^[3]

Notice what has just happened, because it is the quiet beginning of everything this book is about. The eye delivered two pictures, each centred on itself — two ego-centred snapshots. The chiasm does not forward them; it *re-sorts* them, carving the retinal image along a seam that has nothing to do with either eyeball and everything to do with the world: left world here, right world there. The first act of the central visual system is to begin trading a frame anchored to the eye for a frame anchored to the world. We will watch that trade continue, station by station, until — much later, and much higher — it yields a map with no viewer at its centre at all. It starts here, at the X.

Reading the wiring backward

Because the sorting is orderly, it can be read backward, and this is the physician's oldest trick with the visual system: the *pattern* of blindness draws a map of *where* the pathway was cut, with a precision that still seems too good to be true. Sever one optic nerve, before the crossing, and one eye alone goes dark — a monocular loss. Injure the crossing fibres in the belly of the chiasm — most often a pituitary tumour swelling up from the sella beneath it

— and the patient loses precisely the crossing traffic, which carries the nasal retinas, which see the *temporal* fields: both outer edges of the world fall away in a bitemporal hemianopia, as though the patient had been fitted with blinkers and walled in toward the midline.^[4] Cut *behind* the chiasm — in the tract, the radiation, or the cortex — and because each side now carries one entire half-field, the same half of the world vanishes in both eyes together: a homonymous hemianopia.^[5] One lesion, one signature. The geometry of the wiring is legible in the geometry of the loss.

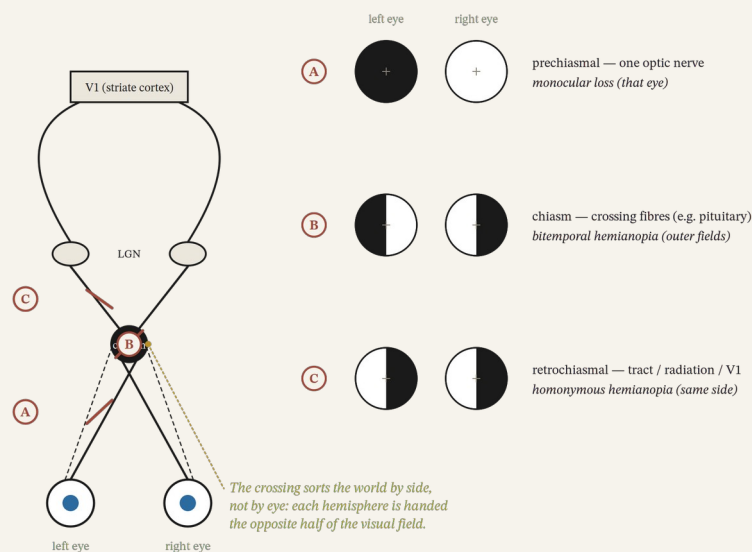


Figure 1. The visual pathway and the geometry of its lesions. Because the crossing is orderly, the site of a lesion — A prechiasmal, B chiasmal, C retrochiasmal — predicts the pattern of field loss.

When the instrument authors the observation

There is a cautionary jewel buried in that map, and it belongs here because it is, in miniature, the epistemology of this whole book. At the junction where an optic nerve joins the chiasm, a very particular defect was long described — blindness in one eye together with a wedge cut from the upper-outer field of the *other* — and for the better part of a century it had a tidy anatomical cause. Examining silver-stained chiasms around the turn of the twentieth century, Wilbrand drew a loop: inferonasal crossing fibres that detour a few millimetres *forward into the opposite optic nerve* before turning back to cross. Damage that loop — "Wilbrand's knee" — and you clip the far eye's upper field. The knee entered the textbooks and lodged there for generations.^[4]

In the normal brain, it is almost certainly not there. In 1997 Horton traced the living pathway in intact monkeys and found no knee whatever — the fibres cross cleanly. The loop appears only *years after an eye has been removed*, when the deafferented half of the chiasm slowly atrophies and shrinks, dragging slack crossing fibres forward into the stump of the dead nerve. Wilbrand had drawn his knee in exactly such material — eyes long lost

before death. The landmark taught for a hundred years was not an anatomy but a scar: an artifact of the very preparation used to find it.^[6] Living confirmation followed — surgeons dividing the nerve at that junction produced no such field defect^[7] — though the case is not perfectly closed; newer light-scattering methods hint at a modest real loop, and the debate flickers on.^[8]

Hold the shape of that story, because we will meet it again and again on the way up. A structure was *seen* — described, drawn, believed, and examined for a century — largely because of how the tissue had been prepared for the seeing. The instrument authored the observation. This is not a lapse of old anatomists to be smiled at from a safe distance; it is the permanent condition of an observer, and it is the first faint sounding of the question this book climbs toward. How much of what is *seen* — here on the microscope slide, and later in the seen world itself — is delivered by the object, and how much is supplied by the act of looking?

Light left at the door

Step back from the threshold. We have crossed a sieve where a wire put on the coat that made it brain; we have watched two eye-centred pictures begin to be re-carved into halves of a world; we have read the wiring off the pattern of its failures. And in none of it has *light* played any part. Light was spent at the door — captured, transduced, and left behind in the first volume. Everything from the lamina cribrosa inward is electricity: spike trains on myelinated wires, sorted by address. The richness that is coming will not live in the fibres, which are only cables, nor in the crossing, which is only a sorting office. It will live in what the brain *does* with the ordering — and the first surprise waits a single synapse ahead, in a small knee of thalamic tissue where, against every intuition, most of the traffic turns out to be running the wrong way: not up from the eye, but *down* from the brain.

2. The lateral geniculate nucleus: the gate that mostly listens to the brain

The little knee

One synapse past the chiasm the optic tract sweeps back and up around the brainstem and ends — most of it — in a small, folded knot of grey matter tucked beneath the thalamus: the *lateral geniculate nucleus*, the "little knee" (Latin *geniculum*), named for the bend in its layered body. The name is a quiet joke on the chapter before, which closed at a knee that was not there; this one is real, and it is where the wrong-way traffic promised at the threshold turns out to run. The tract delivers here almost its whole cargo — in the primate the great majority of ganglion-cell axons end in the geniculate, a smaller stream having peeled off earlier to the midbrain and hypothalamus for the pupil reflex and the body clock — so this is, functionally, *the* gateway from eye to cortex.^[9] It looks the part of a relay: neat, layered, orderly, retinotopic. It is one of the most thoroughly mislabelled structures in the brain.

Six layers, still one eye at a time

Slice it and six layers appear, stacked like the leaves of a book.^[9] The first volume named the retina's three parallel channels — the fast, achromatic **magnocellular** stream from the parasol cells; the fine, chromatic **parvocellular** stream from the midjets; the **koniocellular** stream from the small bistratified cells, carrying blue-yellow — and set them aside as the streams "first hinted at the retina." Here they are hinted no longer but built into the masonry: the two ventral layers are magnocellular, the four dorsal layers parvocellular, and between each pair lies a thin koniocellular leaf.^[9] The parallel channels of Part 1 have become parallel *floors*.

Two features of the stack matter for what follows. First, it is still resolutely **monocular**: each layer is driven by *one* eye alone — they alternate, the ipsilateral eye feeding layers 2, 3 and 5, the contralateral eye feeding 1, 4 and 6 — and the sheets are held in such precise registration that a needle driven straight down through them would pass, at every floor, through the *same point of the visual world*, one eye's view laid exactly over the other's yet not merged with it.^[9] The two eyes have reached the same building and been filed on separate floors; the fusion into a single seen space — binocular vision proper — waits for the cortex. Second, the individual geniculate cell still sees much as a ganglion cell did: a small concentric receptive field, a bright centre and an opposed surround — the centre-surround unit the first volume built in the retina, passed on all but unchanged.^[9] On the evidence of its output alone, the geniculate looks like a faithful repeater: the retina's message, sorted and forwarded.

Then you count what actually lands on a geniculate cell, and the picture turns inside out.

Counting the synapses

Trace every synapse onto a relay neuron in the geniculate — every contact, from every source — and only about **one in twenty, at most one in ten, comes from the retina.**^[10]

The eye, the entire reason the structure exists, supplies five to ten per cent of its input. The rest — the overwhelming majority — arrives from everywhere *but* the eye: a massive descending projection from the visual cortex itself, layer 6 of V1 reaching back down to the very relay that feeds it; inhibitory cells alongside and in the reticular sheet of thalamus draped over the nucleus; and a wash of modulatory fibres climbing from the brainstem — cholinergic, noradrenergic, serotonergic — that carry no image at all, only the state of the animal: drowsy or alert, attending or indifferent.^{[9][10]}

Read the proportion again, because it is one of the quiet scandals of the visual system. The structure set between the eye and the seeing cortex, the supposed relay, spends some *ninety per cent* of its synaptic real estate not listening to the eye. It listens, mostly, to the brain — and above all to the very cortex it is about to report to. The signal has barely entered, and already the destination is talking back down the line.

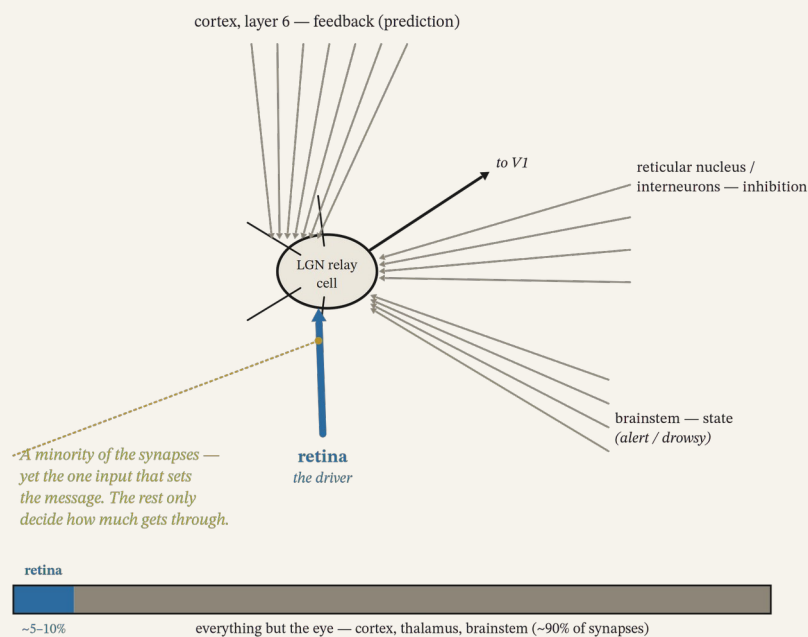


Figure 2. What lands on a geniculate relay cell. The retina is only a small minority of the synapses — the one driver that sets the message — while the great majority is modulatory: cortical feedback, reticular inhibition, and brainstem state.

Driver and modulator

It would be easy to over-read this — to say the eye has been outvoted, that we mostly see what the brain expects and merely glance at the world to confirm it. That overshoots, and the correction runs as deep as the surprise. Sherman and Guillery drew the distinction that rescues the picture: not all synapses are equal, and they fall into two kinds.^[10] A **driver** is an input that *sets the message* — few, powerful, it hands the cell its receptive field and tells it *what* it is about. A **modulator** *adjusts* that message — its gain, its timing, its sharpness, whether it passes crisply or is held back — without changing what the message is *of*. The retinal input, for all that it is a small minority of the synapses, is the **driver**: it alone fixes what a geniculate cell responds to. The vast cortical and brainstem majority are **modulators**: they do not paint the picture, they set the terms on which it is allowed through.

So the true shape of the place is not "the brain overrides the eye." It is subtler and stranger: **the eye's message arrives into a gate already saturated with the brain's expectations and the body's state, and those decide how — and how much — the message is let through.** The retina proposes; the cortex, from above, disposes. The geniculate is the first station in the pathway where seeing is plainly not the passive receipt of what the eye sends, but the brain *conditioning its own input*.

And lest that gate seem a human peculiarity, it is not. The driver-modulator arrangement, and the ninety-per-cent-modulated relay, are a general principle of the mammalian thalamus — holding across mouse, cat and monkey alike, drawn by Sherman and Guillery as a rule of thalamic relays rather than a primate quirk.^[10] The seeing brain's habit of conditioning its own input is a broad solution, not an accident of our lineage. Nor is it the only architecture that reaches sight. Vertebrates route their visual traffic along two great highways — the *thalamofugal*, which mammals swelled into the retina-to-geniculate-to-cortex system traced in these pages, and the *tectofugal*, which runs through the optic tectum (our superior colliculus, the older and more conserved visual centre) and which birds and reptiles lean on instead.^[14] That the mammalian geniculate is one option among several is shown most sharply by the owl: frontally-eyed and binocular as we are, it evolved — along the *other* highway, in a forebrain structure called the Wulst — neurons tuned to orientation and to binocular disparity, topographically mapped, so like the cells of our own visual cortex that they are fairly called its analog, arrived at convergently.^[15] The lesson the chameleon and the radiolarian will carry at the level of the eye and the frame sounds here already at the level of the wiring: the single seen world is authored by *one design among several* the universe allows.

The gate that expects

This is the anatomy the whole ascent has been waiting for. In the nineteenth century Helmholtz proposed that perception is *unconscious inference* — that the brain does not read the world off the senses but *guesses* at the world and uses the senses to check the guess. It was for a long time a philosopher's conjecture with no wiring to stand on. The geniculate's traffic is that wiring made visible: a relay whose dominant input is a descending signal from the cortex, meeting a thin ascending report from the eye. The modern formalisation names the roles exactly — in **predictive coding**, the higher level sends *down* what it expects, and the lower level returns *up* only the **error**, the residue where the world failed to match the guess.^[12] On that reading the geniculate is not relaying an image; it is checking the eye's evidence against the cortex's hypothesis and forwarding the discrepancy. We would not, then, see the retina's picture at all. We would see the brain's best current guess, corrected — only where it must be — by the light.

Honesty demands the flag, and here it is load-bearing rather than decorative. That corticogeniculate feedback exists and dominates numerically is *fact*; that its office is precisely the "prediction" of predictive coding is the **leading interpretation, not a settled result**. The feedback is measured to sharpen receptive fields, tune gain, synchronise the relay's firing, and gate it by attention — that much is data.^{[10][11]} Whether all of it is best read as a single prediction-and-error loop, or as something the theory has yet to capture, is still argued. The anatomy is a fact in search of its exact theory; predictive coding is the most powerful theory we have, and may not be the last.

One measurement pins the direction of travel past dispute. When a person merely *attends* to a region of a scene without moving the eyes, the response of the human geniculate to that region *rises* — the gate opens wider for what the mind has already decided matters, upstream of any conscious sight.^[11] Attention reaches all the way down to the second synapse of the visual system and turns up the gain. The eye holds no opinion about what is important; the geniculate already does, on the brain's instructions.

The stale present

There is a price for all this machinery, and paying it forces the strangest admission of the chapter. Transduction, conduction, synapse upon synapse — the signal that at last reaches awareness is *old*, out of date by some tens of milliseconds, on the order of a tenth of a second. Strictly, we never perceive the present: we perceive a recent past, the world as it stood while the light was still being worked on. And yet a fielder catches the ball and a hand closes on a moving cup; behaviour shows no such lag. The resolution is that the brain does not wait for the news — it **predicts forward**, extrapolating the world's trajectory to render a *now* that cancels its own delay.^[13] The seam shows in the laboratory: flash a spot exactly level with a moving one, and the moving spot is seen *ahead* — the flash-lag effect,

the moving thing carried on into its predicted future while the flash, owning no trajectory to extrapolate, is left where it fired.^[13] (Precisely *what* the illusion proves — forward extrapolation, or a delayed re-reading of the past — is itself contested; that the brain is doing *something* to defeat its own latency is not.)

Set that beside the geniculate's descending traffic and the two facts fold into one. A system whose dominant signal is a top-down expectation, and whose conscious present is rendered ahead of its own data, is not receiving the world — it is *modelling* it, continuously, and sampling the light only often enough to keep the model honest. The present we see is the model's best current hypothesis; the past is what trained it; the light is the correction term. "Now," for a seeing brain, is a construction with a forward lean.

We are two synapses in, and the eye is already the junior partner. What waits one stage further — in the cortex that sends all that expectation down — is where the image is finally broken into its elements and the two eyes' separate floors are at last built into one seen space: the edges, the orientations, the columns of V1, and the first cell that will not fire for a spot of light at all, but only for a line held at the right tilt.

3. V1: the cortex that finds the edges

From the dot to the line

The optic radiation carries the geniculate's output back to the occipital pole, to the primary visual cortex — V1, the striate cortex — the first place the signal becomes *cortex*. And here, at last, the representation is transformed rather than merely relayed. Everything upstream spoke in *dots*: the concentric centre-surround unit, a bright spot ringed by a dark one, the vocabulary the first volume built in the retina and the geniculate passed on all but unchanged. V1 speaks a different language. Its cells will not answer to a spot at all.

Hubel and Wiesel found this, famously, half by accident — a cell in the cat's cortex that ignored every spot they projected, then fired hard as the edge of a glass slide swept across the screen at one particular tilt.^[16] The **simple cell** was the unit they described: it responds to a line, an edge, a bar, but only at a specific *orientation* and only in a specific *place*, built from a row of geniculate centre-surround inputs laid end to end so that an edge at the right angle lights the whole row at once.^[16] Tilt the line off the cell's preferred angle and the response falls away sharply, gone long before the perpendicular. The retina asked *is there light here?*; the simple cell asks *is there an edge, and which way does it lean?*

Beside it sits the **complex cell**, and with it the first breath of abstraction. A complex cell keeps the orientation preference but surrenders the fussiness about position: it fires for a correctly-tilted line *anywhere* in its receptive field, and often for that line *moving*.^{[16][17]} The simple cell reported an edge at a place; the complex cell reports that an edge *exists*, and lets go of exactly where. That small release — orientation kept, position discarded — is the first step of the whole cortical project to come: building descriptions that survive the accidents of the image, the beginning of seeing a *thing* rather than a pattern of light.

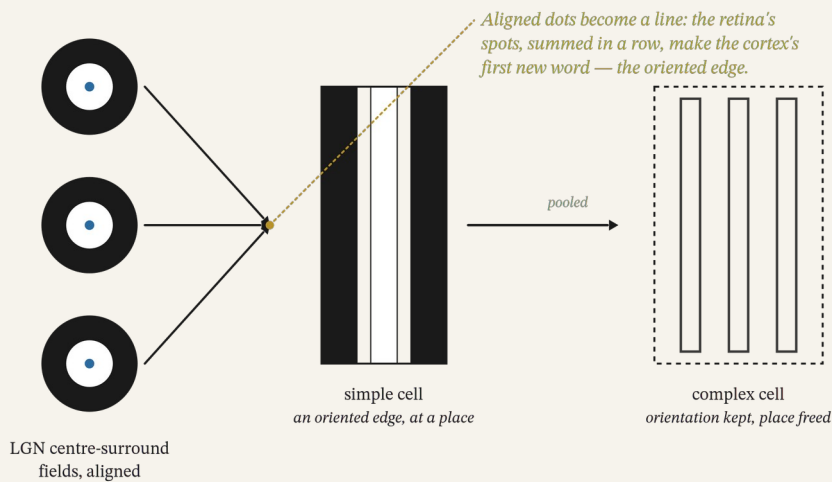


Figure 3. *From the dot to the oriented edge. A row of aligned centre-surround inputs sums into a simple cell tuned to an oriented edge at a place; pooling across position yields the complex cell, which keeps the orientation and lets the place go.*

The two eyes, fused at last

The geniculate kept the two eyes on separate floors; not one of its cells listened to both. In V1 they finally meet. For the first time a single neuron is driven by *both* eyes — a **binocular** cell — and the cortex arranges its cells by which eye leads, into the alternating bands Hubel and Wiesel named **ocular dominance columns**.^[17] More than a merger, this is a new source of information: because the eyes view from slightly apart, a binocular cell can read the small *disparity* between the two images — and disparity is depth. The flat retinal picture, doubled and compared, yields the third dimension. The single, solid, inhabited space we take for the seen world is assembled here, out of two nearly identical flat pictures, by cells that measure how they differ.

That we run a *single* fused viewpoint at all is worth pausing on, because it is a commitment, not a necessity — and the chameleon keeps the other option open. Scanning for prey it holds its two eyes in genuinely separate gaze, and can track *two* small targets at once, one with each eye; only when it has chosen its insect do the eyes swing together into a single binocular fixation, in under a fifth of a second, for the depth its ballistic tongue must have.^[21] Not two wholly independent brains behind two eyes — the popular story, and false; the coordination is real and governed from above — but two viewpoints held apart until one is *needed*, then fused on demand. Our cortex has made the opposite choice: it fuses always, into one Cyclopean world, and buys depth at the price of the second, separate view. The single seen world is not what vision must deliver; it is what *our* brain has committed to.

Columns, hypercolumns, and the magnified map

Step back and the cortex shows a startling order. Drive an electrode straight down through V1 and every cell along the track shares one orientation preference; slide it sideways and the preferred orientation rotates smoothly, degree by degree; slide it the other way and the leading eye alternates.^{[16][17]} Hubel and Wiesel gathered these regularities into the **hypercolumn** — a patch of cortex roughly a millimetre across holding a complete set: every orientation, both eyes, and the colour-processing "blobs," all for one small piece of the visual field. Tile the surface with hypercolumns and you have V1's whole scheme: the same full analysis performed over and over, at every point of a map of visual space.

But the map is wildly uneven, and the unevenness is the point. The central degree or so of vision — the fovea's patch, on the order of a hundredth of a per cent of the retina's area — is given a large fraction of the entire cortex, while the vast periphery is crushed into what remains; ganglion cells near the fovea command several times the cortical territory of peripheral ones.^[18] The first volume's foveal cone density becomes, here, foveal *cortical* real estate — acuity paid for a second time, in neurons. It is the whole system's weighting made cortical: the world is not sampled evenly but lavishly where the eyes point and sparsely everywhere else, and the seamless wide field you seem to possess is stitched over a map that spends almost everything on its centre.

The mathematics of the edge

Why edges? Why should the cortex tear the image into oriented fragments rather than keep it whole? Here the book's claim — that seeing is computation — stops being a metaphor and becomes mathematics. A simple cell's receptive field is, to good approximation, a **Gabor function**: a small grating under a Gaussian window, the mathematical object *optimally localized at once in space and in spatial frequency* — pinning down both where a feature sits and at what scale, as tightly as the two can jointly be known. V1's vocabulary is a wavelet-like decomposition of the image.

And it is not arbitrary. Natural images are hugely redundant — neighbouring points resemble each other, structure repeats — and a nervous system that pays in spikes and energy for all it represents has every reason to *re-code* that redundancy away: the efficient-coding idea Attneave and Barlow proposed half a century ago. Olshausen and Field made it unarguable in 1996. Set a learning system loose on natural images with a single instruction — represent them *sparsely*, as few units active at a time as possible — and the filters that emerge, with no further guidance, are spatially localized, oriented, and bandpass.^[19] They are, in short, simple cells. V1's edge detectors are what an *optimal, sparse code for the natural world looks like*. The cortex did not stumble on edges; edges are the efficient alphabet for the images this world actually contains. Seeing, at its first cortical step, is

already the brain performing — fast, wet, and unawares — the mathematics of compression.

The edge that isn't there

One step further and the construction shows its hand. Take a Kanizsa figure: three notched discs set so that a bright triangle seems to hover above them, its sharp sides plainly visible where in fact there is nothing — no line, no change of luminance at all. Von der Heydt and colleagues put electrodes into the visual cortex of alert monkeys and found cells — in V2, the area just past V1 — that fire for these **illusory contours** exactly as though a real edge crossed their receptive field; weaken the illusion for the observer and the neuron's response weakens with it. The cells of V1 proper, by and large, could not "see" the phantom edges; the drawing-in happens a step later.^[20] The brain is reporting an edge the light never carried — completing, inferring, *drawing the contour it expects* onto the world. In that single cell firing for a line that does not exist is the whole thesis in miniature: seeing is not receipt but construction, the construction is literal, and it begins early.

This is also where the previous chapter's loop closes. The flood of descending prediction that swamps the geniculate — the ninety per cent — originates in this cortex: V1's layer 6 is the source of that feedback, and V1 is itself drenched in feedback from V2 and above, its "simple" responses already bent by the context surrounding each receptive field. Even at the first cortical touch, seeing is not a clean upward flow from eye to mind; it is a conversation, mostly running downward, into which the light is admitted as evidence.

What V1 and its neighbours hand on, then, is no longer a picture. It is a rich, binocular, orientation-parsed, sparsely-coded, context-shaded, foveally-magnified *description* — and still a description of the **image**, not yet of the **world**. Edges are not objects; disparities are not places. The next step divides the labour in two — one stream running up and forward to ask *what* a thing is, another up and back to compute *where* it is and *how* to act on it — and it is on that second road that the eye-anchored frame born at the chiasm will at last be rebuilt into a world you can move through.

4. The two streams: where seeing becomes knowing and acting

The fork

Past V1 the single description does not stay single. It splits into two great cortical roads, and the split is one of the deepest facts about how brains see. One stream runs *downward and forward*, into the temporal lobe — the **ventral** stream, carrying the fine, coloured, detailed parvocellular signal toward the work of recognition: *what* a thing is. The other runs *upward and back*, into the parietal lobe — the **dorsal** stream, fed heavily by the fast magnocellular channel the first volume named, toward the work of location and action: *where* a thing is, and *how* to lay a hand on it.^[22]

Ungerleider and Mishkin drew the division in 1982 as *what* versus *where*.^[22] A decade later Goodale and Milner sharpened it, and the sharpening matters: the dorsal stream, they argued, is better understood not as *where* but as *how* — vision in the service of action, computing the size, shape, orientation and distance a movement needs, fast and largely beneath awareness, then throwing them away. The ventral stream is vision for *perception* — the slow, conscious, enduring seen world you can recognise, name, and remember.^[23] Two visions, running in parallel from the same eyes: one you experience, and one, mostly, you do not.

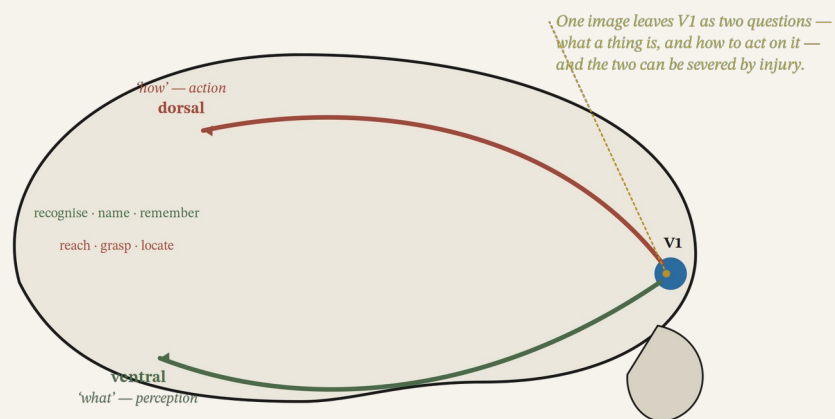


Figure 4. The two cortical streams. One image leaves V1 as two questions — the ventral ‘what’ of perception, bound for the temporal lobe, and the dorsal ‘how’ of action, bound for the parietal — and the two can be severed by injury.

Knowing without doing, doing without knowing

That the two can be *surgically separated* is what turned a tidy diagram into a profound claim, and the evidence is a set of patients in whom seeing has come apart along its seam.

Patient D.F., after carbon-monoxide poisoning injured her ventral stream, was left with **visual form agnosia**: shown a slot cut at some angle, she could not say which way it tilted, could not match it, could not report its orientation at all. Yet handed a card and asked to *post* it through that same slot, she turned her wrist to the exact angle and slipped it in without hesitation.^[23] Her hand knew what her sight could not tell her. The acting vision was intact; only the knowing vision was gone.

Damage the dorsal stream instead — as in the **optic ataxia** of Bálint's syndrome — and the dissociation runs the other way: the patient recognises the cup perfectly, names it, describes it, then gropes for it, the hand sliding past, unable to shape the grasp the object plainly calls for. Knowing without doing; doing without knowing. The same visual world, split down the middle by which half of the brain is hurt.

And then the sharpest case of all. Destroy V1 itself and the corresponding half of the visual field goes *blind* — no sight, no experience of anything there. Yet ask such a patient to *guess* where a light is, or whether a bar in the dark field moves left or right, and the guesses come back well above chance: pointing, detecting, discriminating, with no awareness of seeing at all. Weiskrantz named it **blindsight**, and it is carried by the ancient routes that bypass the cortex — the superior colliculus, the pulvinar — the visual brain beneath the seeing one.^[24] Here is vision with the seer removed: the eye still guides the hand, and no one is home to watch. Hold that; it is the trailhead of the last chapter.

Rebuilding the frame

The dorsal stream's other great labour is spatial, and it is the one promised all the way back at the chiasm. The eye delivers a picture centred on itself; but to *act* you need a thing's position not on your retina but relative to your hand, your head, the room. The parietal cortex performs exactly these conversions — eye-centred into head-centred into body-centred into world-centred — the coordinate transforms that turn a smear of light on a curved retina into a reachable *place*. Here the eye-anchored frame born at the X is at last rebuilt into a world you can move through.

And that frame is *authored* — provably, by a famous demonstration. Put on goggles that invert the visual field, as George Stratton did in 1897 and the Innsbruck experimenters did for weeks together, and the world is at first an unusable, upside-down chaos; but over days of living in it, coherent action returns — reaching and walking come right, the disorder subsides.^[25] The naïve story, that the image "flips back upright," is almost certainly wrong, and the truth is better: the retinal image never changes at all. What re-authors itself is the

sensorimotor mapping — the brain relearns which action yields which consequence, and "up" quietly redefines itself as *the arrangement in which reaching and walking succeed*. Held and Hein pinned the crucial condition: it is **active** movement that drives the adaptation; a subject moved passively through the very same views adapts far less.^[26] You must *act* in the world to re-author your frame of it. (Whether the scene ever truly *looks* upright again — what the goggles do to conscious perception — remains debated; that behaviour realigns is not.)

There is one boundary condition, though, that the brain does *not* author from scratch, and it deserves naming precisely, because it is the seen world's single fixed peg. Up and down are not vision's to invent; they are imported — from gravity, sensed by the vestibular otoliths, and merely *referenced* by the eyes. When those cues conflict, the vertical becomes a visible negotiation: in the rod-and-frame effect, a tilted surrounding frame drags a person's sense of vertical away from true, toward the frame — yet gravity hauls it partway back, so the perceived upright is a weighted compromise between what the eyes see and what the inner ear feels.^[27] Strip gravity away and the peg pulls loose: astronauts in orbit, their otoliths unloaded, lose a stable "down" altogether — floors become ceilings, the visual world reorients without warning. So the scalpel cut is this. *Spatial* seeing is gravity-**polarised** — up and down borrowed from the otoliths — while left/right and front/back are *body*-polarised, hung off the direction the animal faces. The frame is very nearly all self-made — the prism goggles prove it — but it is nailed, at one corner, to the one external reference physics hands over for free.

The map with no viewer

Follow the rebuild to its summit and you reach something the first chapter promised. Beyond the parietal transforms, in the hippocampus and the cortex feeding it, are cells that have let go of the viewpoint entirely. A **place cell** fires when the animal stands in one particular spot of a room, and keeps firing for that spot no matter which way the animal faces.^[28] Beside it, **grid cells** tile the whole floor with a regular hexagonal lattice of firing — a coordinate mesh laid over the world itself.^[29] Together they encode position not as *bearing-from-me* but as *location-in-the-world*: an **allocentric** map, a chart of places and their relations that survives when you turn your head, shut your eyes, or leave the room. This is the achievement flagged at the chiasm, where the brain first began re-sorting the eye's picture by the world rather than by the eyeball — here completed, a map with no viewer at its centre at all. The two ego-centred snapshots the eyes delivered have become a world one can hold in mind and move through in the dark.

What the streams leave us

Step back from the fork and its two roads. Seeing has turned out to be not one thing but several, running at once: a slow conscious vision that knows *what*, and a fast unconscious

vision that does *how*; a frame the brain authors from action and pins, at a single corner, to gravity; an ascent from two flat eye-centred snapshots to a viewer-free map of the world. At every stage the through-line has held — the richness lives in the computation — and the world we seem simply to *receive* has in fact been built, sorted, predicted, completed, and re-framed, station by station, since the moment the signal crossed the Stargate.

Which leaves the two questions the whole climb has been walking toward, and which no amount of wiring has yet touched. If the seen world is authored — from the light, but not *given* by it — then what is that authored world, and how far does the authoring reach? And, harder and last: a blindsighted brain guides the hand with no one watching, a dorsal stream acts in the dark — so why is the *rest* of seeing lit from within? Why is there something it is *like* to see the river, and not merely to compute it? The final chapter takes up the seen world, and the seer.

5. Coda: the authored world and the seer

The world with the light off

We began by following the light to the back of the eye, and we have climbed, station by station, until the light was left behind. It is worth saying plainly how far. Once seeing became a *computation* — a running estimate built by the cortex and merely corrected by the retina — it was, in principle, freed from the retina altogether; and the brain takes the liberty. Close your eyes and picture a face: the visual cortex stirs, and a face is there, unlit. Remember a room, and you walk through it in the dark. And each night, in dreaming, the whole apparatus runs *open-loop* — the generative model firing a complete world, vivid and inhabited, with the sensory gates shut and no correction arriving from anywhere (§2). The seen world does not need the world. That is the strongest possible statement of the through-line, and it is not a metaphor: it is a thing the sleeping brain does every night.

The dreams of the blind prove the claim from its far edge. People blind from birth have no visual imagery in their dreams at all; they dream in sound and touch and taste and smell — the senses they own — and their eyes, having no visual scene to track, do not even make the sleeping movements that in the sighted betray a dream.^[30] Those who lose sight later keep their visual dreams, which fade across the decades; the dividing line falls near the age of five to seven, the window in which the visual model was built.^[30] The lesson is exact: dream imagery is the generative model running, and a model can generate only what experience taught it. No light ever admitted, no visual world to author — awake or asleep. What is inherited is the *machinery* of world-building, not its contents: the projector crosses the generations, the film is shot fresh in each life.

And the machinery is *supramodal* — it does not care which sense feeds it, or whether any does. The congenitally blind build a spatial world regardless, out of touch and hearing, in the very cortex vision would have used. Synesthesia (a sense running *extra*), sensory substitution (a sense running *in place of* vision), and dreaming (the senses running on *nothing*) are three faces of one fact: the seen world is a construction that will take any input, or none. It is made — and we know it is made because we have watched it run with the world switched off.

The shadow and the fixed point

If it is made, then of *what*, and from what? Here the ascent turns to face the oldest question about it, and a plain observation reorganises everything: **no sense delivers the object.** Vision least of all. You have never seen a river; you have seen *light that grazed a river* and carried off its signature — a reflection, a shadow — and "river" is merely the best hypothesis your brain holds for what would throw such a shadow. Touch seems to escape

this: the object itself presses your skin, at zero remove. But even that press is transduced into spikes and *interpreted*; the felt shape, too, is a reading. Three senses, three distances of mediation, and one object that none of them contains — because the object is not in any single proxy. It is the *invariant* the proxies agree on, the thing that best explains why the light, the sound, and the touch all covary. This is why two senses are worth so much more than twice one: combine vision and touch and the brain weights each by its reliability and fuses them into an estimate more precise than either alone — objectivity assembled, statistically, out of agreement.^[31] The real is not handed over; it is inferred as the common cause of the shadows, the fixed point they all orbit.

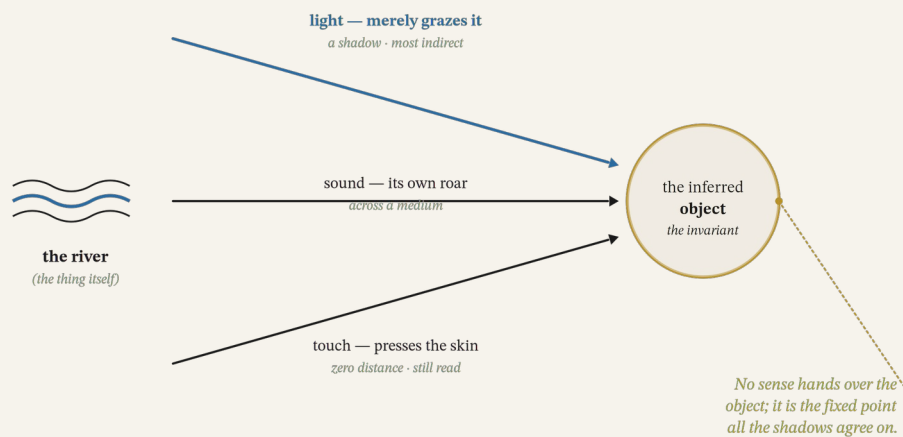


Figure 5. *The ladder of mediation. No sense delivers the object: light merely grazes it, sound is its own doing across a medium, touch presses at zero distance — and the object is the invariant on which all three proxies agree.*

It is Plato's cave, rebuilt on an optics bench. The prisoners take the shadows on the wall for the things themselves; the retinal image is that wall. The one cruel difference is that Plato's prisoner could in principle turn around, and we cannot: the shadow is the only thing any eye has ever received. There is no facing the object — only inferring it better.

Why a world, and not a proof

Which raises the question this whole book has circled, now sharp enough to state: if the world is structure, and the brain an inference engine working from evidence, why does it hand us a *seen world* — colour, depth, a lit scene — rather than a proof, a conclusion, a set of coordinates? Why *see* at all, instead of merely *inferring*?

The answer is that the dichotomy is false, and its falseness is the thesis. **Seeing is not the alternative to inference; seeing is inference** — Helmholtz's unconscious conclusion, compiled into fast, wet, parallel hardware and *rendered* as a navigable display. The

prediction-and-error machinery met at the geniculate (§2) is exactly this inference, made of neurons. And it cannot be replaced by pure logic, for three solid reasons. Logic needs premises, and the shadows are the only evidence there is — perception is how the premises get *in*. Deduction is serial and brittle and drowns in the combinatorics of a real scene, where a fast, approximate estimator catches the ball. And the brain has no direct access to the world's structure, only to projected samples of it — so perception must be the *running estimation of that structure from the shadows*, rendered into the one format a body can act on: space, time, and objects laid out to hand. Kant said it first — that space and time are the mind's own forms for making the world surveyable; the modern and sharper version is the interface, the icon on the screen that is not the file and need not resemble it, only let you act.^[32] On this reading, "seeing instead of inferring" is a category mistake: *seeing is the interface to an inference*, the cheapest control surface for a thing that must reach and dodge and grasp before it can finish thinking.

Honesty requires the standing rival, and it is a serious one. Gibson's ecological view denies the whole picture: the light filling a room already *specifies*, lawfully and completely, the layout that produced it, so perception is not inference but *direct pickup* of information that was there all along — no inner shadow, nothing to reconstruct.^[33] On that account the river's light is not a shadow standing *for* the river but information *about* it, fully present, waiting to be resonated with. Whether seeing is inference or direct pickup is a genuine and open fight; the book stages it and does not pretend to end it. But if the inferential picture holds, the conclusion is vertiginous and clean: seeing is the brain doing empirical mathematics, far too fast and too far below awareness to feel like mathematics — and it *feels like a world* because a world is the interface mathematics wears when a body has to live inside it.

Forms all the way down?

Grant that, and the ground gives way beneath it — the last and sharpest turn. If space and time are the mind's forms, is *mathematics* itself one of them, and so itself flexible? Kant thought geometry simply *was* the science of space and arithmetic the science of time, which made mathematics as much a form of the mind as the intuitions it was built from. History then cashed the bet in the hardest way: Gauss and Bolyai and Lobachevsky and Riemann built consistent geometries in which Kant's "necessary" parallel postulate fails, and General Relativity found that physical space is in fact one of them — curved. The form Kant swore the mind could not escape turned out to be one option among many, and not the one the universe chose. And yet — the subtlety on which everything balances — each of those geometries is internally *rigid*: "in this space the angles fall short of two right angles" is as fixed, as mind-indifferent, as any theorem Euclid proved. What flexed was *which* structure describes our world; what did not flex was the structure of each structure. The plurality is a plurality of *rigid* things.

So the fault line is whether that rigidity is *found* or *made* — whether mathematics is discovered, mind-independent, the deepest description of any possible reality (the Platonist's least-flexible thing in the universe), or constructed, a formal game whose rules we choose (and even logic, it turns out, comes in plural kinds). The book cannot settle a question that has outlasted every mind that raised it. But it can see where the question goes, because it bites its own tail. We said seeing is an inference that estimates a mathematical reality. But if mathematics is *also* a form the mind imposes, then that "mathematical reality" is itself part of the interface — and it is **forms all the way down**, an authored world resting on an authored floor resting on nothing, no thing-in-itself ever reached. There is exactly one exit from the regress, and it is the Platonist one: that mathematics alone among the forms is *discovery* and not imposition — the single fixed floor, the noumenon that simply *is* mathematical structure. Kant's unreachable thing-in-itself, set against a universe made of nothing but mathematics.^[34] This is the anchor question in its largest form — *is there a floor under the seen world, or is authorship all the way down?* — and even the humblest object keeps the mystery alive: what, after all, is zero — the mark we invented to name *nothing*, without which almost no mathematics runs? Even the void, it seems, had to be authored before it could be found.

The culture has built its own instruments for standing at this edge, and the book may nod to them where an argument wants a face: the dream given architecture and stacked dream-within-dream until no level is certainly the waking one; the rendered world that is real *somewhere* but not *here*; the simulation that discovers it contains simulations, and then that it is one. Seasoning, not scaffolding — the modern descendants of the cave and the demon, useful because they let a reader *feel* in the body an argument millennia older than any film.

The seer who could be otherwise

There is one more move the book has been saving, and it is the cleanest proof that the frame lives in the *seer* and not in the light. Imagine a mind with no body like ours — a neutrally-buoyant, spherically-symmetric sensor with no gravity to pull an "up" and no face to fix a "front." It could not build the ego-centred world we take for granted, for it has no *centre*; it would have to perceive *allocentrically* from the start — the world as a web of relations among things, sensed over the whole sphere at once, its natural mathematics not perspective but the harmonics of a sphere, its space free of the occlusion and vanishing-point geometry that define ours. Feed it the very same photons and pressure waves, and it authors a wholly different seen world — not because the physics changed, but because the *author* did. That is the thesis at its highest altitude: swap the seer, and the same light becomes another world. (And a smaller cousin of the puzzle sits inside our own skull: sever the two hemispheres and each can behave as though it sees on its own — so how many seers is one brain, and what ordinarily makes them one? We walk up to that, too, and no further.)

Notice, finally, where the anchor question quietly closes. *Why electricity, and not light?* Because to author a world at all, the computation must be **medium-independent** — unmoored from whatever knocked on the door, able to run on spikes whether they came from an eye, an ear, a fingertip, or from nothing but the model itself in the dark. Light gets the data in; electricity makes the sense; and it is precisely the medium-independence of the sense-making that lets the seen world be run generatively, corrected by the light or floated free of it. The medium question and the meaning question were the same question all along.

The seer

And so the ascent arrives where it has always been heading, and stops. Grant everything the climb has established: that seeing is inference; that the inference is rendered as an interface; that it estimates a world it can never directly touch, from shadows, medium-blind, the frame supplied by the seer; that it runs even with the light switched off. Grant all of it — and one thing remains entirely unexplained, the one thing the book was really about. A blindsighted brain guides the hand to a target it does not experience seeing; the dorsal stream acts in the dark; a pure inference engine could compute *obstacle, wet, roaring* and light up no scene, feel nothing, be no one. Why, then, is the rest of seeing *lit from within*? Why is there something it is *like* to see the river — the silver on the water, that particular blue — and not merely to compute it?

That is the hard problem, and it is the summit this staircase climbs to.^[35] Everything below it we can answer, and this book has tried to: how the light is caught, sorted, predicted, parsed, completed, re-framed, and raised into a world. The last step — from the seen world to the *seer*, from the inference to the felt fact of undergoing it — we cannot answer, and the one discipline this book has held from its first page forbids it to pretend otherwise. So we do the only honest thing at a true cliff: we walk all the way up to it, and stop.

The first volume ended at the last structure that is still the eye, and said: *what lies beyond is the brain learning to see*. This one has followed the seeing to its end — through the Stargate, past the gate that expects, through the edge that isn't there, the two roads, and the map with no viewer — and arrives at the last structure that is still explicable, to say: *what lies beyond is the seer, learning that it sees*. We do not know what that is. But we have earned the right to stand here, in the lit world we made from a handful of shadows, and to find the not-knowing not empty, but full.

References

1. Neuroanatomy, Cranial Nerve 2 (Optic). In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2022. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK507907/>
2. Neuroanatomy, Visual Pathway. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2026. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK553189/>
3. Kupfer C, Chumbley L, Downer JC. Quantitative histology of optic nerve, optic tract and lateral geniculate nucleus of man. *J Anat*. 1967;101(Pt 3):393–401.
4. Neuroanatomy, Optic Chiasm. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2023. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK542287/>
5. Zhang X, Kedar S, Lynn MJ, Newman NJ, Biouesse V. Homonymous hemianopias: clinical–anatomic correlations in 904 cases. *Neurology*. 2006;66(6):906–910.
6. Horton JC. Wilbrand's knee of the primate optic chiasm is an artefact of monocular enucleation. *Trans Am Ophthalmol Soc*. 1997;95:579–609.
7. Lee JH, Tobias S, Kwon JT, Sade B, Kosmorsky G. Wilbrand's knee: does it exist? *Surg Neurol*. 2006;66(1):11–17.
8. Shin RK, Qureshi RA, Harris NR, et al. Wilbrand knee. *Neurology*. 2014;82(5):459–460.
9. Neuroanatomy, Nucleus Lateral Geniculate. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2023. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK541137/>
10. Sherman SM, Guillery RW. The role of the thalamus in the flow of information to the cortex. *Philos Trans R Soc Lond B Biol Sci*. 2002;357(1428):1695–1708.
11. O'Connor DH, Fukui MM, Pinsk MA, Kastner S. Attention modulates responses in the human lateral geniculate nucleus. *Nat Neurosci*. 2002;5(11):1203–1209.
12. Rao RPN, Ballard DH. Predictive coding in the visual cortex: a functional interpretation of some extra-classical receptive-field effects. *Nat Neurosci*. 1999;2(1):79–87.
13. Nijhawan R. Motion extrapolation in catching. *Nature*. 1994;370(6487):256–257.
14. Wylie DR, Gutiérrez-Ibáñez C, Pakan JMP, Iwaniuk AN. The optic tectum of birds: mapping our way to understanding visual processing. *Can J Exp Psychol*. 2009;63(4):328–338.
15. Pettigrew JD, Konishi M. Neurons selective for orientation and binocular disparity in the visual Wulst of the barn owl (*Tyto alba*). *Science*. 1976;193(4254):675–678.
16. Hubel DH, Wiesel TN. Receptive fields, binocular interaction and functional architecture in the cat's visual cortex. *J Physiol*. 1962;160(1):106–154.
17. Hubel DH, Wiesel TN. Receptive fields and functional architecture of monkey striate cortex. *J Physiol*. 1968;195(1):215–243.
18. Daniel PM, Whitteridge D. The representation of the visual field on the cerebral cortex in monkeys. *J Physiol*. 1961;159(2):203–221.

19. Olshausen BA, Field DJ. Emergence of simple-cell receptive field properties by learning a sparse code for natural images. *Nature*. 1996;381(6583):607–609.
20. von der Heydt R, Peterhans E, Baumgartner G. Illusory contours and cortical neuron responses. *Science*. 1984;224(4654):1260–1262.
21. Ketter Katz H, Lustig A, Lev-Ari T, Nov Y, Rivlin E, Katzir G. Eye movements in chameleons are not truly independent — evidence from simultaneous monocular tracking of two targets. *J Exp Biol*. 2015;218(13):2097–2105.
22. Ungerleider LG, Mishkin M. Two cortical visual systems. In: Ingle DJ, Goodale MA, Mansfield RJW, editors. *Analysis of Visual Behavior*. Cambridge (MA): MIT Press; 1982. p. 549–586.
23. Goodale MA, Milner AD. Separate visual pathways for perception and action. *Trends Neurosci*. 1992;15(1):20–25.
24. Weiskrantz L, Warrington EK, Sanders MD, Marshall J. Visual capacity in the hemianopic field following a restricted occipital ablation. *Brain*. 1974;97(4):709–728.
25. Stratton GM. Vision without inversion of the retinal image. *Psychol Rev*. 1897;4(4):341–360; 4(5):463–481.
26. Held R, Hein A. Movement-produced stimulation in the development of visually guided behavior. *J Comp Physiol Psychol*. 1963;56(5):872–876.
27. Witkin HA, Asch SE. Studies in space orientation. IV. Further experiments on perception of the upright with displaced visual fields. *J Exp Psychol*. 1948;38(6):762–782.
28. O'Keefe J, Dostrovsky J. The hippocampus as a spatial map. Preliminary evidence from unit activity in the freely-moving rat. *Brain Res*. 1971;34(1):171–175.
29. Hafting T, Fyhn M, Molden S, Moser MB, Moser EI. Microstructure of a spatial map in the entorhinal cortex. *Nature*. 2005;436(7052):801–806.
30. Meaidi A, Jennum P, Ptito M, Kupers R. The sensory construction of dreams and nightmare frequency in congenitally blind and late blind individuals. *Sleep Med*. 2014;15(5):586–595.
31. Ernst MO, Banks MS. Humans integrate visual and haptic information in a statistically optimal fashion. *Nature*. 2002;415(6870):429–433.
32. Hoffman DD. *The Case Against Reality: Why Evolution Hid the Truth from Our Eyes*. New York: W. W. Norton; 2019.
33. Gibson JJ. *The Ecological Approach to Visual Perception*. Boston: Houghton Mifflin; 1979.
34. Tegmark M. *Our Mathematical Universe: My Quest for the Ultimate Nature of Reality*. New York: Alfred A. Knopf; 2014.
35. Chalmers DJ. Facing up to the problem of consciousness. *J Conscious Stud*. 1995;2(3):200–219.